Unit 3: Measurement — Challenge Pack

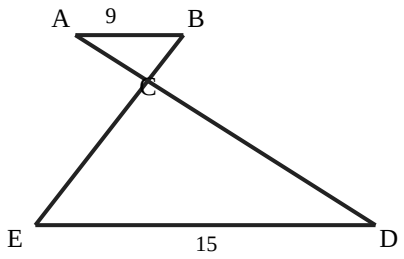
Math 9E · Similar Triangles · Unit Conversions · Surface Area · Volume & Capacity · Scale & Isometric Drawings · Trigonometry review

How to use: Part A has very hard questions for each lesson. Part B has the challenging questions your teacher is most likely to ask, based on the patterns in your worksheets (the ★★★ and ★★★★ problems, the "Challenge before next class" and the proofs). Parts D and E add more worksheet-style and predicted questions. Worked answers for everything are in Part C at the end. ★★★ = hard, ★★★★ = very hard.

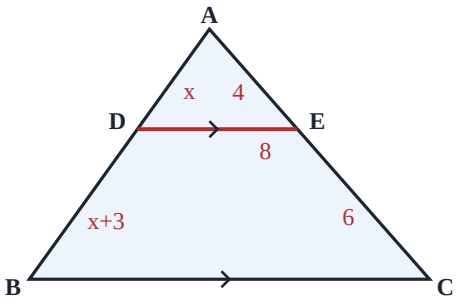
PART A — Very Hard Questions by Lesson

A1. Similar Triangles & Thales' Theorem (Lesson 3.1)

A1.1 ★★★ Segments AD and BE cross at C (an "hourglass"). AB ∥ ED, AB = 9 cm, ED = 15 cm and AD = 20 cm. Find AC and CD.

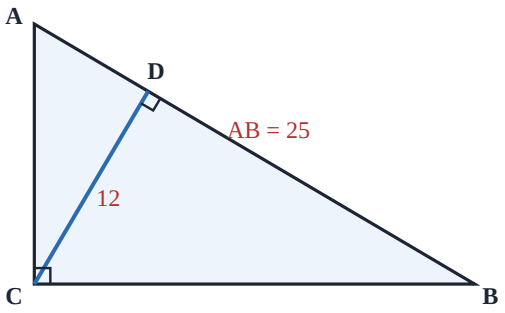


A1.2 ★★★ In ΔABC, DE ∥ BC with D on AB and E on AC. AD = x, DB = x + 3, AE = 4 and EC = 6. (a) Find x. (b) If DE = 8, find BC.

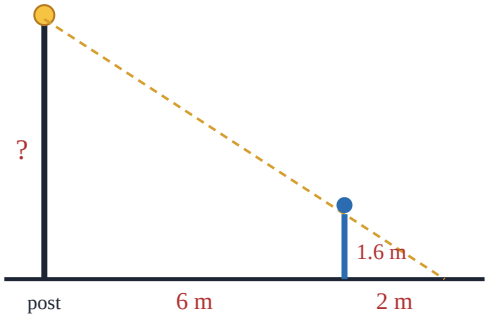


A1.3 ★★★ Two similar triangles have perimeters 24 cm and 40 cm. The larger triangle has an area of 75 cm². Find the area of the smaller triangle.

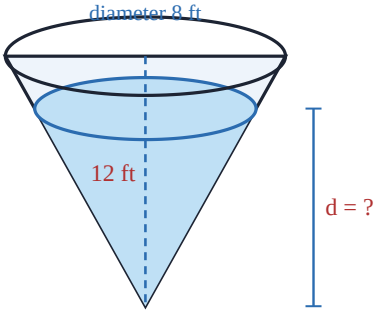
A1.4 ★★★★ ΔABC has a right angle at C. CD is the altitude to the hypotenuse, AB = 25 and CD = 12. Find AD and DB (AD < DB), then the legs AC and BC, the perimeter and the area of ΔABC.



A1.5 ★★★★ A 1.6 m student stands 6 m from a lamp post and casts a 2 m shadow. (a) How tall is the lamp post? (b) The student walks 2 m farther away from the post. How long is the new shadow?



A1.6 ★★★★ A conical tank (point down) is 12 ft deep and the diameter of its top is 8 ft. (a) To what depth must it be filled so that it holds exactly **half** its capacity? (b) What is the area of the water surface at that depth?



A1.7 ★★★★ **Proof.** In the hourglass figure of A1.1 (AB ∥ ED), prove that ΔABC ∼ ΔDEC, and then prove that AC · CE = BC · CD.

A2. Unit Conversions (Lesson 3.2)

A2.1 ★★★ A farm has an area of 2.5 km². Convert this to (a) m², (b) hectares (1 ha = 10 000 m²), (c) acres.

A2.2 ★★★ Convert 7.2×10^{-4} km to µm, and 0.003 Gm to mm. Write both answers in scientific notation.

A2.3 ★★★ A car travels at 90 km/h. Convert this to (a) m/s, (b) miles per hour.

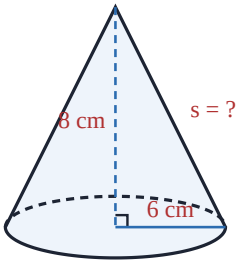
A2.4 ★★★★ 1 ft = 30.48 cm. (a) How many cm² are in 1 ft²? (b) A room is 12 ft × 15 ft. Find its area in m², to the nearest hundredth.

A2.5 ★★★★ A doctor orders 15 mg of medicine per kg of body mass for a child with a mass of 44 lb (1 kg = 2.2 lb). The syrup label reads 250 mg in 5 mL. How many mL should the nurse give?

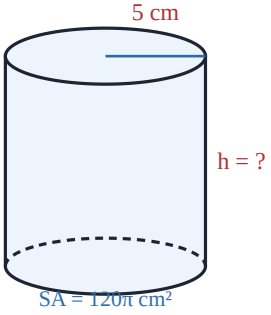
A2.6 ★★★★ Convert 3.2 m³ to cm³ and to litres. Explain why 1 m³ ≠ 100 cm³.

A3. Surface Area (Lesson 3.2)

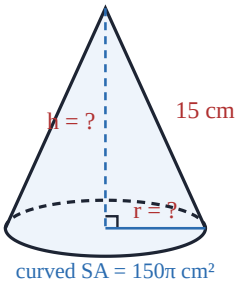
A3.1 ★★★ A cone has radius 6 cm and **height** 8 cm (not slant height). Find its total surface area in terms of π and to the nearest tenth.



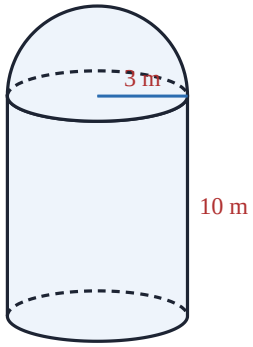
A3.2 ★★★ A closed cylinder has radius 5 cm and total surface area 120π cm². Find its height.



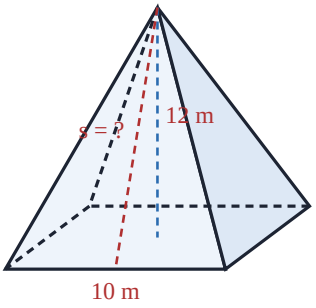
A3.3 ★★★★ A cone has a curved surface area of 150π cm² and a slant height of 15 cm. Find the radius, then the height to the nearest hundredth.



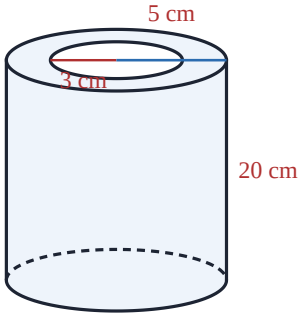
A3.4 ★★★★ A grain silo is a cylinder (radius 3 m, height 10 m) topped by a hemisphere. Only the outside walls and the dome are painted (not the base). One can of paint covers 12 m² and costs $38. Find the area painted, the number of cans and the total cost with 12% tax.



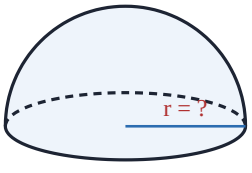
A3.5 ★★★ A right square pyramid has base side 10 m and height 12 m. Find its slant height and total surface area.



A3.6 ★★★★ A pipe is 20 cm long with outer radius 5 cm and inner radius 3 cm. Find the total surface area: outside, inside and both ring-shaped ends.



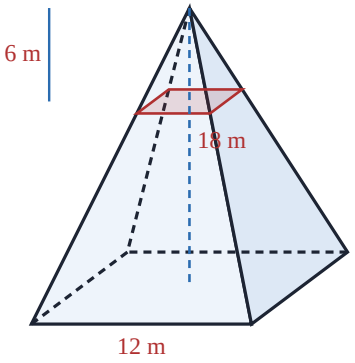
A3.7 ★★★ A solid hemisphere has total surface area 75π cm². Find its radius. (Remember the flat face!)



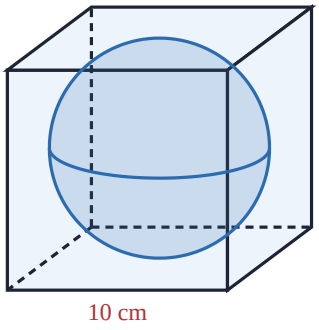
solid hemisphere

A4. Volume & Capacity (Lesson 3.3)

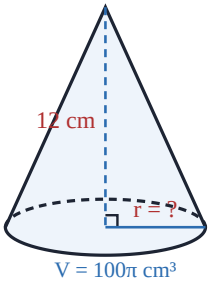
A4.1 ★★★★ A square pyramid has base 12 m × 12 m and height 18 m. It is cut horizontally 6 m from the apex and the top is removed. Find the volume of the remaining piece (frustum).



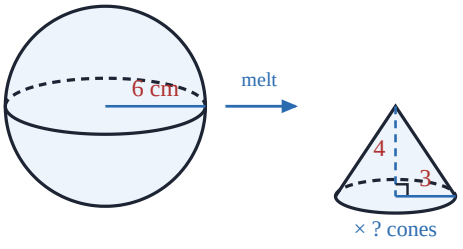
A4.2 ★★★ A sphere fits exactly inside a cube of side 10 cm. Find the volume of empty space and the percentage of the cube that is empty.



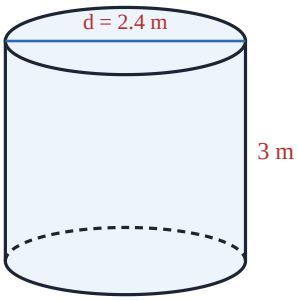
A4.3 ★★★★ A cone has volume 100π cm³ and height 12 cm. Find its radius, slant height and total surface area.



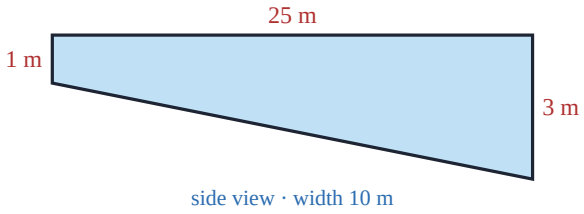
A4.4 ★★★★ A metal sphere of radius 6 cm is melted and recast into small cones with radius 3 cm and height 4 cm. How many cones can be made?



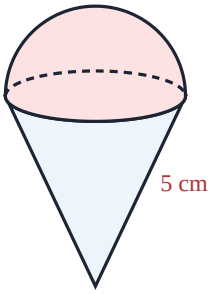
A4.5 ★★★ A cylindrical tank has diameter 2.4 m and height 3 m. (a) Find its capacity in kL. (b) A pump fills it at 40 L/min. How many hours will it take?



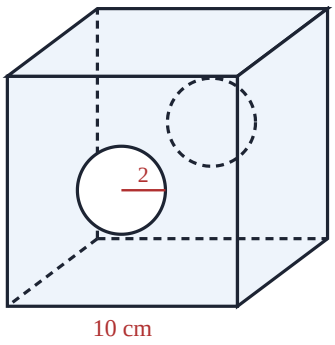
A4.6 ★★★★ A pool is 25 m long and 10 m wide. Its depth slopes evenly from 1 m at one end to 3 m at the other end. Find its capacity in litres. How many days does it take to fill with a hose delivering 20 L/min?



A4.7 ★★★ An ice-cream cone has slant height 5 cm and radius 3 cm, topped by a hemisphere of ice cream of radius 3 cm. Find the total volume.



A4.8 ★★★★ A cube of side 10 cm has a cylindrical hole of radius 2 cm drilled straight through it, from the centre of one face to the centre of the opposite face. Find the remaining volume and the new total surface area.



A5. Scale, Scale Diagrams & Isometric Drawings (Lesson 3.4)

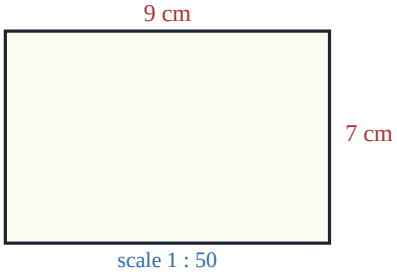
A5.1 ★★★★ A map has scale 1:25 000. (a) Two towns are 18 cm apart on the map. How far apart are they in km? (b) A park covers 12 cm² on the map. Find its actual area in m² and in hectares. (Careful: area does not use the same factor!)

A5.2 ★★★★ A model car is built at a scale of 1:18. The model's fuel tank holds 9 mL. How many litres does the real tank hold?

A5.3 ★★★ A real wall is 4.2 m long and is drawn as 3.5 cm on a plan. Write the scale as 1 : n. Is it a reduction or an enlargement?

A5.4 ★★★ A cell 6 µm wide appears 2.4 cm wide in a textbook image. Find the scale and the scale factor.

A5.5 ★★★★ A floor plan has scale 1:50. A room measures 9 cm × 7 cm on the plan. Flooring costs \$42/m². Find the total cost with 12% tax.



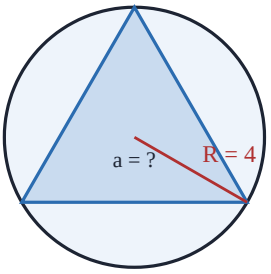
A5.6 ★★★★ A solid is built from 1-cm cubes. The numbers on its top-view grid show how many cubes are stacked on each square (the back row is at the top).

3	**2**	**1**
2	**1**	
1		

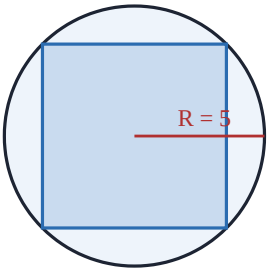
(a) How many cubes are there? (b) Describe or draw the front view and the right view. (c) Find the total surface area of the solid.

A6. Trigonometry & Inscribed Prisms (review of earlier problems)

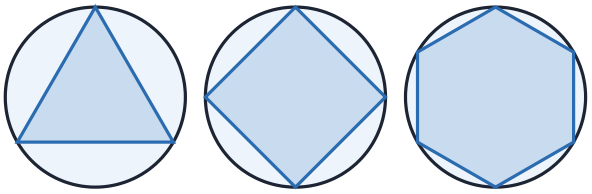
A6.1 ★★★★ The largest possible equilateral triangular prism is cut from a cylinder of radius 4 cm and length 10 cm. Use trigonometry to find the side of the triangle, then find the wasted volume and the percentage wasted.



A6.2 ★★★★ The largest square prism is cut from a cylinder of radius 5 cm and length 8 cm. Find the side of the square and the wasted volume.



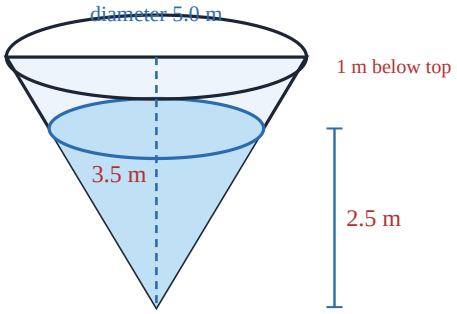
A6.3 ★★★★ Compare the percentage of the cylinder that is used when the inscribed prism has a triangular, a square or a regular hexagonal cross-section. Which wastes the least? Explain why.



PART B — Predicted Challenging Test Questions

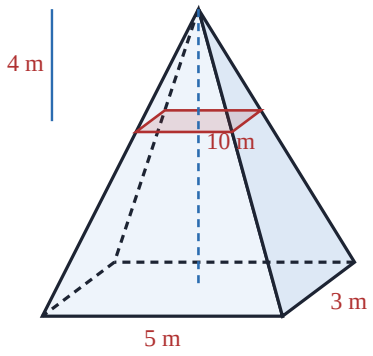
These questions combine skills the same way your worksheets do. The yellow note explains why each is likely.

B1 An underground tank is a right cone with its apex pointing down. Base diameter 5.0 m, height 3.5 m. (a) Find its capacity in kL to the nearest tenth. (b) How much water is in it when the water level is 1 m below the top? (c) What percentage full is it?



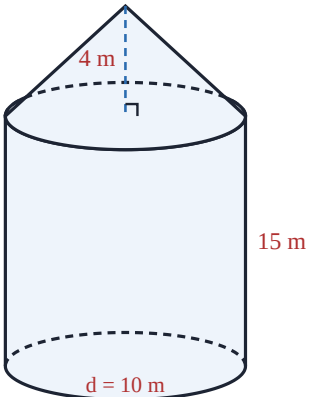
Why: Practice 3.3 #4 is a ★★★+ question. Teachers often reuse it with the "1 m below" twist, which needs similar cones (k^3).

B2 A right rectangular pyramid has base 5 m × 3 m and height 10 m. A horizontal cut is made 4 m from the apex and the top pyramid is removed. Find the volume of the remaining piece.



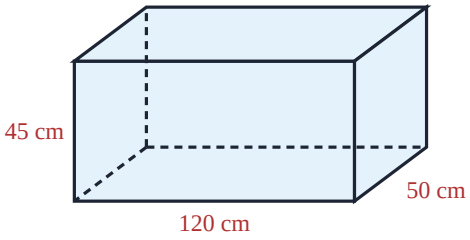
Why: Practice 3.3 #9 (★★★+) with different numbers.

B3 A silo is a cylinder with a conical roof. Diameter 10 m, cylinder height 15 m, roof height 4 m. The walls and roof need sealant (not the base). One bucket covers 30 m² and costs $65. Find the area (2 decimal places), number of buckets and total cost.



Why: Practice 3.2 #8. You must find the slant height with Pythagoras first, which is where most students lose marks.

B4 A fish tank is 120 cm × 50 cm × 45 cm. Fancy goldfish need at least 75 US gallons (1 gal = 3.785 L). (a) Is the tank big enough? (b) Keeping length and width the same, what height would hold exactly 75 gallons?



Why: Practice 3.3 #6, extended with a reverse step.

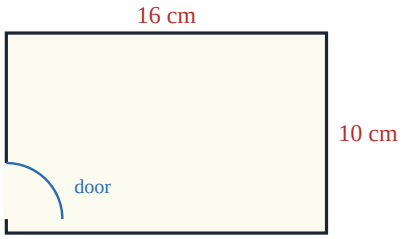
B5 A doctor orders 0.6 g of syrup medicine. The label reads 250 mg in 5 mL. How many mL should be given? How many teaspoons is that (1 tsp = 5 mL)?

Why: Practice 3.2 #3–#4. The trap is the g → mg conversion.

B6 Two similar cones have volumes 54π cm³ and 250π cm³. (a) Find the ratio of their radii. (b) The small cone has surface area 36π cm². Find the surface area of the large cone.

Why: Practice 3.1 #7 (★★★+) used k^2 for areas. A test question may go one step further to k^3.

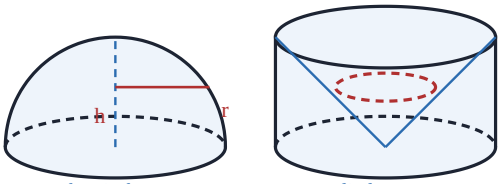
B7 A bathroom is drawn on grid paper where 4 cm represents 1 m. On the plan it is 16 cm × 10 cm. The walls are 3 m tall and have one door of 0.9 m × 2.1 m that is not painted. Tiling costs \$225/m² and painting costs \$16/m². Find the total cost including 12% tax.



4 cm represents 1 m

Why: Practice 3.4 #8 (★★★★). It combines scale, area, perimeter and cost.

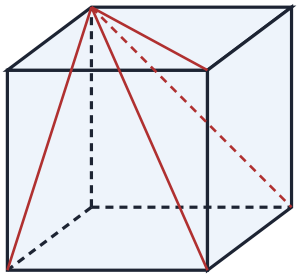
B8 **Derivation.** Use Cavalieri's principle with a hemisphere and a cylinder with a cone removed (same radius r and height r) to derive $V_{sphere} = (4/3)\pi r^3$.



hemisphere cylinder − cone

Why: Practice 3.3 #13 (★★★★). Proofs like this are very common as the "challenge" question on a test.

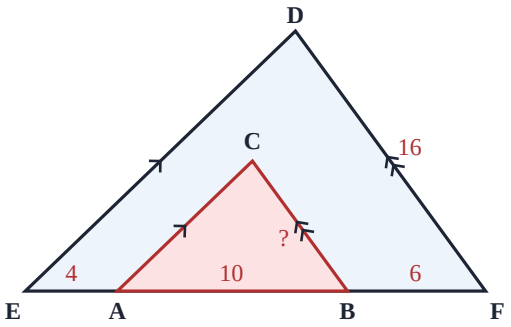
B9 **Derivation.** Explain why the volume of a pyramid is one-third of the volume of a prism with the same base and height.



3 pyramids share one corner

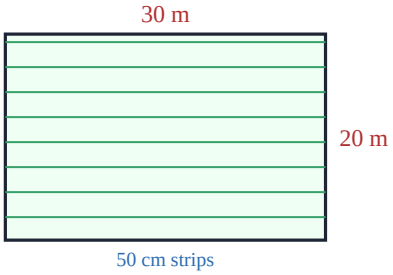
Why: it was the "Challenge before next class" on your answer-key page.

B10 In ΔEFD, points A and B lie on EF with EA = 4, AB = 10 and BF = 6. Point C is inside the triangle with AC ∥ ED and BC ∥ FD. FD = 16. (a) Prove ΔABC ∼ ΔEFD. (b) Find BC.



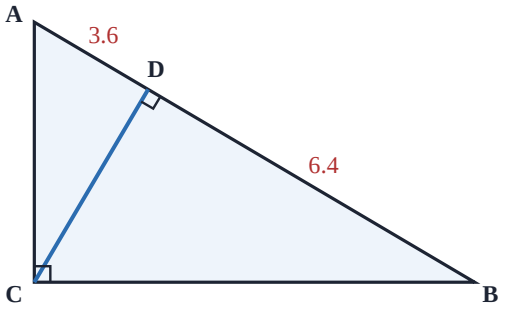
Why: Practice 3.1 #9 asks for a similarity proof plus finding x.

B11 A rectangular lawn is 30 m × 20 m. A mower cuts a 50-cm strip and mows back and forth along the 30 m length. Find the distance travelled in km and the time in minutes at 4 km/h.



Why: Practice 3.2 #9 (★★★★), a simpler shape with the same reasoning.

B12 In right ΔABC (right angle at C), CD ⊥ AB. Prove $CD^2 = AD \cdot DB$. Then, if AD = 3.6 and DB = 6.4, find CD, AC and BC.



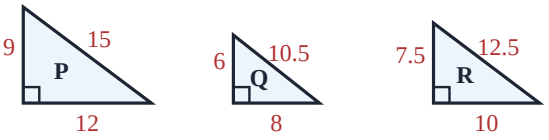
Why: you were asked to prove this in class. Expect a proof followed by numbers.

PART D — More Very Hard Questions (Worksheet-Style, Lessons 3.1–3.3)

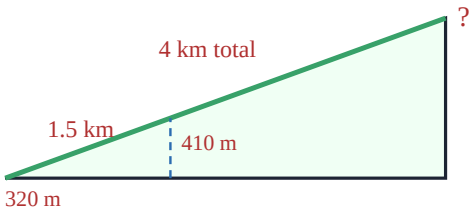
These follow the exact question types in Practice 3.1, 3.2 and 3.3, with new numbers and extra steps.

D1. Similar Triangles

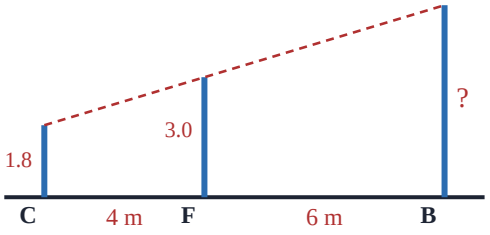
D1.1 ★★★ Triangle P has sides 9, 12, 15. Triangle Q has sides 6, 8, 10.5. Triangle R has sides 7.5, 10, 12.5. Which triangle(s) are similar to P? Show the ratios.



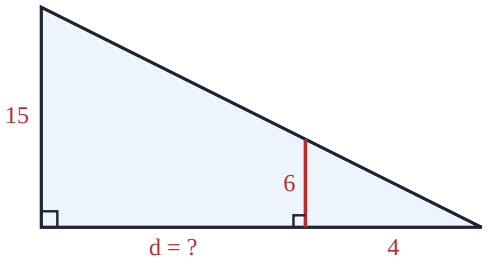
D1.2 ★★★ A hiker starts at an elevation of 320 m. After walking 1.5 km up a straight, evenly sloped trail, the elevation is 410 m. The whole trail is 4 km long. What is the elevation at the top and the total gain? What are you assuming?



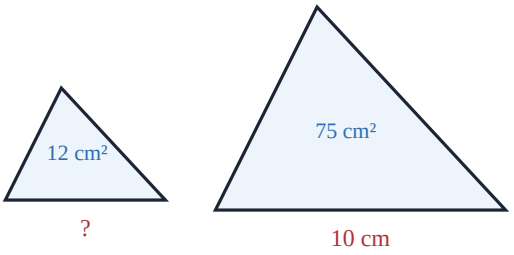
D1.3 ★★★★ Three vertical posts stand at points C, F and B on level ground, with CF = 4 m and FB = 6 m. The post at C is 1.8 m tall and the post at F is 3.0 m tall. The tops of all three posts lie on one straight line. How tall is the post at B? (Hint: subtract 1.8 m to make similar triangles.)



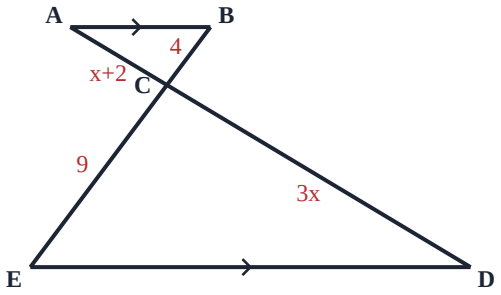
D1.4 ★★★ A large right triangle has a vertical leg of 15. Inside it, a smaller similar right triangle shares the same bottom corner and has a vertical leg of 6 and a base of 4. The big triangle's base is 4 + d. Find d.



D1.5 ★★★★ Two similar triangles have areas 12 cm² and 75 cm². The shortest side of the large triangle is 10 cm. Find the shortest side of the small triangle.



D1.6 ★★★★ In an hourglass figure, AD and BE cross at C and AB ∥ ED. AC = x + 2, CD = 3x, BC = 4 and CE = 9. Find x, AC and CD.



D2. Unit Conversions

D2.1 ★★★ Convert 2.5 miles to metres.

D2.2 ★★★★ A field is 3 acres. Convert it to hectares and to m² (use 1 ha = 2.47 acres).

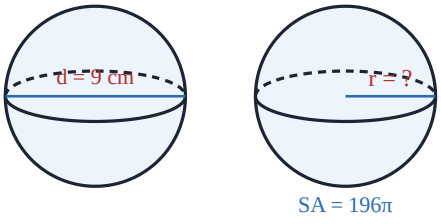
D2.3 ★★★ Convert 0.000045 Tm to km in scientific notation.

D2.4 ★★★★ (a) A doctor orders 2 g of Ampicillin per day, in 4 equal doses. The capsules are 250 mg. How many capsules per dose and per day? (b) 1 L of saline must drip in over 8 hours. What is the rate in mL per minute?

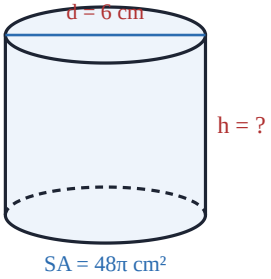
D2.5 ★★★★ How many litres are in 1 ft³? (1 ft = 30.48 cm)

D3. Surface Area

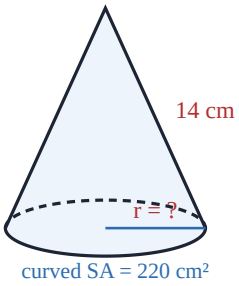
D3.1 ★★★ (a) Find the surface area of a sphere with diameter 9 cm. (b) A different sphere has surface area 196π cm². Find its radius.



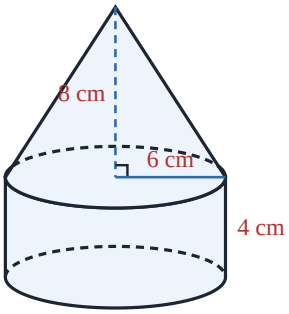
D3.2 ★★★ A closed cylinder has diameter 6 cm and surface area 48π cm². Find its height.



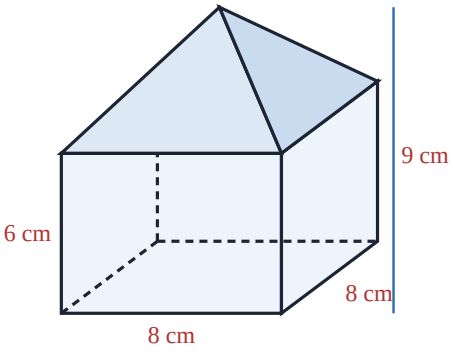
D3.3 ★★★ A cone has curved surface area 220 cm² and slant height 14 cm. Find r to 1 decimal place.



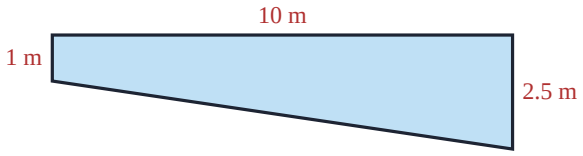
D3.4 ★★★★ A cone sits on top of a cylinder, both with radius 6 cm. The cylinder is 4 cm tall and the cone is 8 cm tall (height, not slant). Find the total outside surface area, including the bottom.



D3.5 ★★★★ A square pyramid sits on top of a box with an 8 cm × 8 cm base and walls 6 cm tall. The total height of the shape is 9 cm. Find the total surface area.



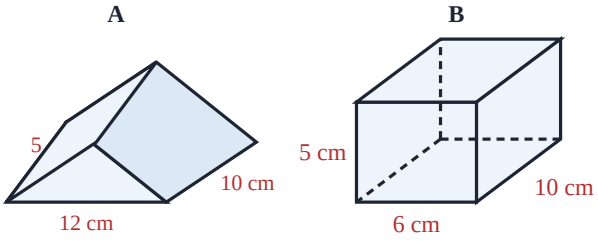
D3.6 ★★★★ A pool is 10 m long and 4 m wide. The depth slopes from 1 m to 2.5 m. Tile the bottom, both side walls and both end walls. (a) Find the area to be tiled. (b) Tiles cost \$40 USD/m² and 1 USD = 1.38 CAD. Find the cost in CAD.



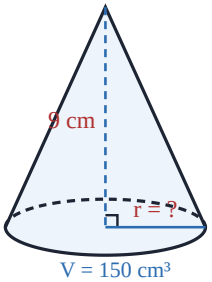
side view · width 4 m

D4. Volume & Capacity

D4.1 ★★★ Prism A has a triangular cross-section (base 12 cm, height 5 cm) and is 10 cm long. Prism B is 6 cm × 5 cm × 10 cm. Carl says they cannot have the same volume because the shapes are different. Is he right? Use Cavalieri's principle.

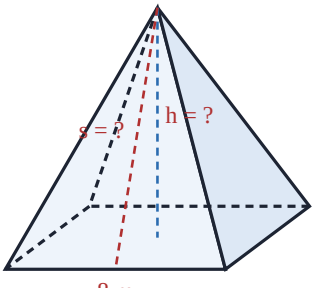


D4.2 ★★★ A cone has volume 150 cm³ and height 9 cm. Find its radius to the nearest hundredth.



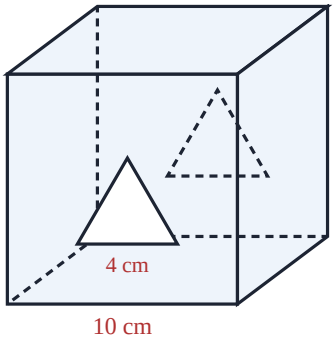
V = 150 cm³

D4.3 ★★★★ A right square pyramid has volume 256 m³ and base side 8 m. Find its height and its slant height to the nearest hundredth.

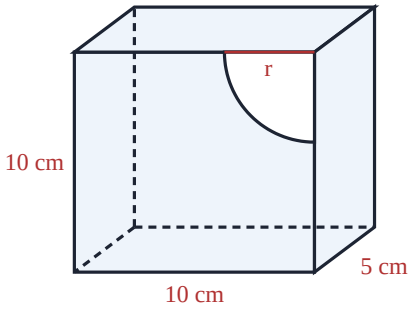


V = 256 m³

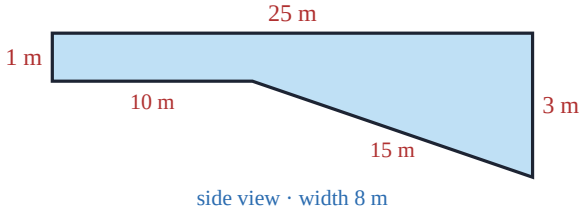
D4.4 ★★★ An equilateral triangular prism hole (side 4 cm) is cut all the way through a 10 cm cube. Find the remaining volume.



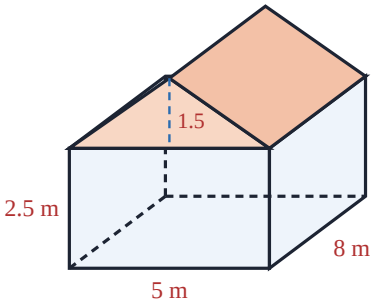
D4.5 ★★★★ A quarter cylinder is cut from one edge of a 10 cm × 10 cm × 5 cm cuboid. The quarter cylinder runs the 5 cm length. The remaining volume is 400 cm³. Find the radius.



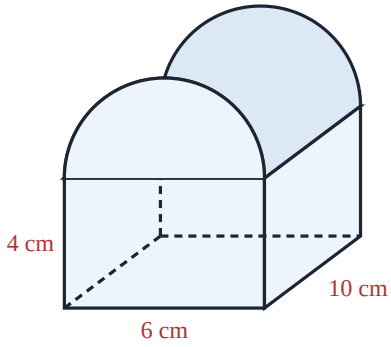
D4.6 ★★★★ A pool is 8 m wide. The first 10 m is a flat shallow end 1 m deep. Then the floor slopes evenly over 15 m to a depth of 3 m. Find the capacity in litres.



D4.7 ★★★ A shed is a box 5 m wide, 8 m long and 2.5 m tall with a triangular roof prism 1.5 m high on top. Find its volume.

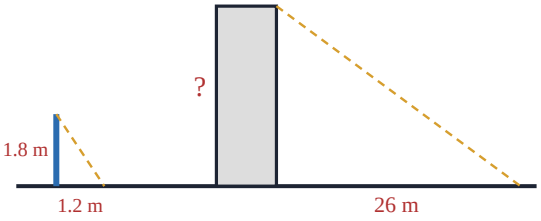


D4.8 ★★★ A box 6 cm × 10 cm × 4 cm has a half-cylinder of diameter 6 cm on top, running the 10 cm length. Find the volume to the nearest hundredth.



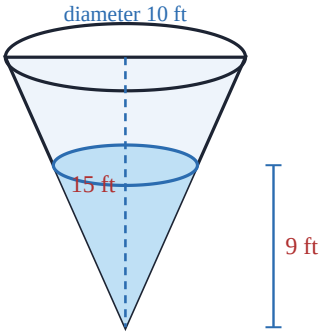
PART E — More Predicted Test Questions

E1 A 1.8 m pole casts a 1.2 m shadow. At the same time a building casts a 26 m shadow. (a) Find the building's height. (b) Later, the building's shadow is 39 m. How long is the pole's shadow then?



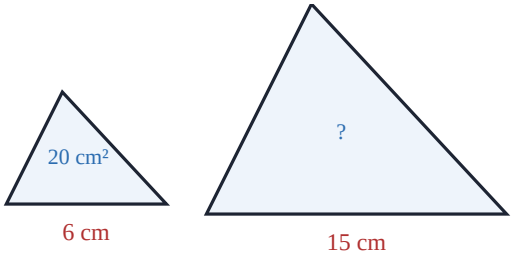
Why: Delbert/flagpole problem (3.1 Ex. 3 and #4) with a second step.

E2 A conical tank (point down) is 15 ft deep with top diameter 10 ft. It is filled to a depth of 9 ft. Find (a) the area of the water surface and (b) the volume of water.



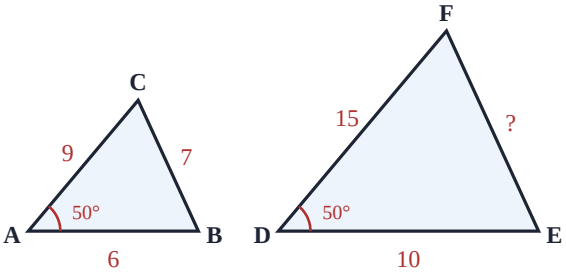
Why: Practice 3.1 #5, with volume added.

E3 Corresponding sides of two similar triangles are 6 cm and 15 cm. The small triangle has area 20 cm². Find the area of the large one.



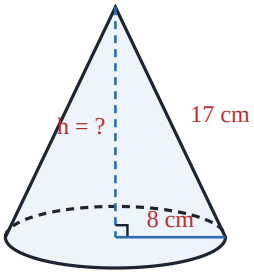
Why: Practice 3.1 #7 in reverse.

E4 In ΔABC, AB = 6, AC = 9 and ∠A = 50°. In ΔDEF, DE = 10, DF = 15 and ∠D = 50°. (a) Prove the triangles are similar. (b) If BC = 7, find EF.



Why: SAS similarity is in your notes and is easy to test as a proof.

E5 A cone has radius 8 cm and slant height 17 cm. Find its height, volume and total surface area.



Why: it links Pythagoras, volume and surface area in one question.

PART C — Answers & Worked Solutions

A1. Similar Triangles

A1.1 AB ∥ ED ⇒ ΔABC ~ ΔDEC (AA: vertical angles at C, alternate angles). AC/CD = 9/15 = 3/5. AC = 20 · 3/8 = **7.5 cm**, CD = **12.5 cm**.

A1.2 (a) x/(x+3) = 4/6 ⇒ 6x = 4x + 12 ⇒ **x = 6**. (b) AB = 15. DE/BC = AD/AB = 6/15 ⇒ BC = 8 · 15/6 = **20**.

A1.3 k = 24/40 = 0.6, so the area ratio is $k^2 = 0.36$. Small area = 75 × 0.36 = **27 cm²**.

A1.4 AD · DB = 144 and AD + DB = 25 ⇒ $t^2 - 25t + 144 = 0$ ⇒ **AD = 9, DB = 16**. $AC^2 = 9 \cdot 25 = 225$ ⇒ **AC = 15**. $BC^2 = 16 \cdot 25 = 400$ ⇒ **BC = 20**. Perimeter = **60**. Area = ½ · 15 · 20 = **150** (check: ½ · 25 · 12 = 150 ✓).

A1.5 (a) The shadow tip is 6 + 2 = 8 m from the post. H/8 = 1.6/2 ⇒ **H = 6.4 m**. (b) Now the student is 8 m away: 1.6/s = 6.4/(8 + s) ⇒ 6.4s = 12.8 + 1.6s ⇒ 4.8s = 12.8 ⇒ **s ≈ 2.67 m**.

A1.6 (a) The water forms a cone similar to the tank. Volume ratio $k^3 = 1/2$ ⇒ $k = 1/\sqrt[3]{2} \approx 0.7937$. Depth = 12 · 0.7937 ≈ **9.52 ft** (not 6 ft!). (b) r = 4 · 0.7937 ≈ 3.175 ft. Area = $\pi r^2 \approx$ **31.67 ft²**.

A1.7 ∠ACB = ∠DCE (vertical angles). ∠BAC = ∠EDC (alternate angles, AB ∥ ED). So ΔABC ~ ΔDEC (AA). Corresponding sides: AC/DC = BC/EC. Cross-multiply: **AC · EC = BC · DC** ∎

A2. Unit Conversions

A2.1 1 km² = (1000 m)² = 10^6 m². (a) **2 500 000 m²** (b) ÷ 10 000 = **250 ha** (c) × 2.47 = **617.5 acres**.

A2.2 7.2×10^{-4} km = 0.72 m = **7.2×10^5 µm**. 0.003 Gm = 3×10^6 m = **3×10^9 mm**.

A2.3 (a) 90 × 1000 ÷ 3600 = **25 m/s** (b) 90 ÷ 1.609 ≈ **55.9 mph**.

A2.4 (a) $30.48^2 \approx$ **929.03 cm²** (b) 180 ft² × 0.3048^2 m² ≈ **16.72 m²**.

A2.5 44 ÷ 2.2 = 20 kg. Dose = 15 × 20 = 300 mg. Volume = 300 × 5/250 = **6 mL**.

A2.6 1 m³ = 100 cm × 100 cm × 100 cm = 10^6 cm³. 3.2 m³ = **3 200 000 cm³ = 3200 L**. You must cube the factor because there are three dimensions.

A3. Surface Area

A3.1 $s = \sqrt{(6^2 + 8^2)} = 10$. SA = π(6)(10) + π($6^2$) = **96π ≈ 301.6 cm²**.

A3.2 2π(25) + 2π(5)h = 120π ⇒ 50 + 10h = 120 ⇒ **h = 7 cm**.

A3.3 πr(15) = 150π ⇒ **r = 10 cm**. $h = \sqrt{(15^2 - 10^2)} = \sqrt{125} \approx$ **11.18 cm**.

A3.4 Walls 2π(3)(10) = 60π. Dome = ½ · 4π(3^2) = 18π. Total = 78π ≈ **245.04 m²**. 245.04 ÷ 12 = 20.4 ⇒ **21 cans** (always round up). 21 × 38 = $798; with tax ×1.12 = **$893.76**.

A3.5 Slant = $\sqrt{(12^2 + 5^2)}$ = **13 m**. SA = 10^2 + 4 · ½ · 10 · 13 = 100 + 260 = **360 m²**.

A3.6 Outside 2π(5)(20) = 200π; inside 2π(3)(20) = 120π; ends 2 · π($5^2 - 3^2$) = 32π. Total = **352π ≈ 1105.8 cm²**.

A3.7 Curved part $2\pi r^2$ + flat circle $\pi r^2 = 3\pi r^2 = 75\pi$ ⇒ $r^2 = 25$ ⇒ **r = 5 cm**.

A4. Volume & Capacity

A4.1 $V_{big} = \frac{1}{3} \cdot 144 \cdot 18 = 864$ m³. $k = 6/18 = 1/3 \Rightarrow V_{small} = 864/27 = 32$ m³ (check: base 4 × 4, $\frac{1}{3} \cdot 16 \cdot 6 = 32$ ✓). Remaining = **832 m³**.

A4.2 Sphere r = 5: $V = (4/3)\pi(125) \approx 523.60$. Empty = 1000 − 523.60 ≈ **476.40 cm³ ≈ 47.6%**.

A4.3 $\frac{1}{3}\pi r^2(12) = 100\pi \Rightarrow r^2 = 25 \Rightarrow$ **r = 5**. $s = \sqrt{(25 + 144)} =$ **13**. $SA = \pi(5)(13) + 25\pi =$ **90π ≈ 282.7 cm²**.

A4.4 $V_{sphere} = (4/3)\pi(216) = 288\pi$. $V_{cone} = \frac{1}{3}\pi(9)(4) = 12\pi$. $288\pi \div 12\pi =$ **24 cones**.

A4.5 (a) $V = \pi(1.2^2)(3) \approx 13.57$ m³ ≈ **13.6 kL**. (b) 13 572 L ÷ 40 ≈ 339.3 min ≈ **5.65 h**.

A4.6 The side face is a trapezoid: (1 + 3)/2 × 25 = 50 m². V = 50 × 10 = 500 m³ = **500 000 L**. Time = 25 000 min ≈ **17.4 days**.

A4.7 $h = \sqrt{(5^2 - 3^2)} = 4$. Cone $\frac{1}{3}\pi(9)(4) = 12\pi$; hemisphere $(2/3)\pi(27) = 18\pi$. Total = **30π ≈ 94.25 cm³**.

A4.8 $V = 1000 - \pi(2^2)(10) = 1000 - 40\pi \approx$ **874.34 cm³**. $SA = 600 - 2(4\pi) + 2\pi(2)(10) = 600 + 32\pi \approx$ **700.53 cm²**. The SA *increases* because the inside wall of the hole is added.

A5. Scale & Isometric

A5.1 (a) 18 × 25 000 = 450 000 cm = **4.5 km**. (b) Area factor = $25\,000^2 = 6.25 \times 10^8$. 12 cm² × $6.25 \times 10^8 = 7.5 \times 10^9$ cm² = **750 000 m² = 75 ha**.

A5.2 Volume factor = $18^3 = 5832$. 9 mL × 5832 = 52 488 mL ≈ **52.5 L**.

A5.3 3.5 cm : 420 cm = **1 : 120**, a **reduction**.

A5.4 2.4 cm = 24 000 µm. 24 000 : 6 = **4000 : 1** (enlargement). Scale factor = **4000**.

A5.5 Room 4.5 m × 3.5 m = 15.75 m². 15.75 × 42 = \$661.50; × 1.12 = **\$740.88**.

A5.6 (a) 3+2+1+2+1+1 = **10 cubes**. (b) Front view: columns of heights 3, 2, 1 from left to right (a staircase). Right view: heights 1, 2, 3 from left to right (front row on the left). Top view: an L-shaped staircase of 6 squares. (c) Top + bottom = 2 × 6 = 12; front + back = 2 × 6 = 12; left + right = 2 × 6 = 12 ⇒ **36 cm²**.

A6. Trigonometry & Inscribed Prisms

A6.1 From $\cos 30° = (a/2)/R$: $a = 2R \cos 30° = R\sqrt{3} =$ **4√3 ≈ 6.93 cm**. $V_{cyl} = 160\pi \approx 502.65$. $V_{prism} = (\sqrt{3}/4)(48)(10) = 120\sqrt{3} \approx 207.85$. Waste ≈ **294.81 cm³ ≈ 58.7%**.

A6.2 The diagonal of the square is the diameter 2R ⇒ side = $R\sqrt{2} =$ **5√2 ≈ 7.07 cm**. Square area = $2R^2 = 50$. Waste = $200\pi - 400 = 200(\pi - 2) \approx$ **228.32 cm³** (36.3%).

A6.3 Used: triangle $3\sqrt{3}/(4\pi) \approx$ **41.3%**; square $2/\pi \approx$ **63.7%**; hexagon $3\sqrt{3}/(2\pi) \approx$ **82.7%**. The hexagon wastes the least. The more sides the polygon has, the closer it fills the circle.

Part B — Predicted Questions

B1 (a) $V = \frac{1}{3}\pi(2.5^2)(3.5) \approx 22.91$ m³ ≈ **22.9 kL**. (b) Water depth = 2.5 m ⇒ $k = 2.5/3.5 = 5/7$. $V = 22.91 \times (5/7)^3 \approx$ **8.3 kL**. (c) $(5/7)^3 \approx$ **36.4%**. Being only 1 m below the top still leaves the tank well under half full!

B2 $V_{big} = \frac{1}{3}(15)(10) = 50$ m³. $k = 4/10 = 0.4 \Rightarrow V_{small} = 50 \times 0.064 = 3.2$ m³. Remaining = **46.8 m³**.

B3 r = 5. Roof slant = $\sqrt{(5^2 + 4^2)} = \sqrt{41} \approx 6.403$. Walls $2\pi(5)(15) = 150\pi$; roof $\pi(5)(6.403) \approx 32.02\pi$. Total ≈ **571.82 m²**. 571.82 ÷ 30 = 19.06 ⇒ **20 buckets ⇒ $1300**.

B4 (a) 120 × 50 × 45 = 270 000 cm³ = 270 L ÷ 3.785 ≈ 71.3 gal. **No, it is not big enough.** (b) 75 × 3.785 = 283.875 L = 283 875 cm³. h = 283 875 ÷ (120 × 50) ≈ **47.3 cm**.

B5 0.6 g = 600 mg. 600 × 5/250 = **12 mL = 2.4 tsp**.

B6 (a) $k^3 = 250/54 = 125/27$ ⇒ k = **5/3**, so radii are in the ratio 3 : 5. (b) Area ratio $k^2 = 25/9$ ⇒ $36\pi \times 25/9 = \mathbf{100\pi\ cm^2}$.

B7 Room 4 m × 2.5 m. Floor = 10 m² × $225 = $2250. Walls: perimeter 13 m × 3 m = 39 m² − 1.89 m² (door) = 37.11 m² × $16 = $593.76. Subtotal $2843.76; with tax ×1.12 = **$3185.01**.

B8 Slice both shapes at height h. Hemisphere slice: radius $\sqrt{(r^2 - h^2)}$ (Pythagoras), so area = $\pi(r^2 - h^2)$. Washer: the outer radius is r. The cone (apex at the bottom, height r, radius r) has radius h at height h, so washer area = $\pi r^2 - \pi h^2 = \pi(r^2 - h^2)$. The areas are equal at every height, so by Cavalieri $V_{hemisphere} = V_{cyl} - V_{cone} = \pi r^3 - \frac{1}{3}\pi r^3 = \frac{2}{3}\pi r^3$. Sphere = $2 \times \frac{2}{3}\pi r^3 = \mathbf{(4/3)\pi r^3}$ ∎

B9 A cube of side s can be split into **3 identical pyramids** that share one corner as their apex. Each has a square base s × s and height s, so $V = s^3/3 = \frac{1}{3}$(base area)(height). Stretching the cube into a box (or using Cavalieri) keeps the ratio 1/3, so $V_{pyramid} = \mathbf{\frac{1}{3}Bh}$.

B10 (a) AC ∥ ED ⇒ ∠CAB = ∠DEF (corresponding angles). BC ∥ FD ⇒ ∠CBA = ∠DFE. So ΔABC ∼ ΔEFD (AA). (b) EF = 4 + 10 + 6 = 20, k = AB/EF = 10/20 = 1/2 ⇒ BC = ½ · FD = **8**.

B11 Rows = 20 ÷ 0.5 = 40. Distance = 40 × 30 = 1200 m = **1.2 km**. Time = 1.2 ÷ 4 = 0.3 h = **18 min**.

B12 Proof: ∠A = ∠DCB (both equal 90° − ∠ACD) and both triangles have a right angle at D ⇒ ΔADC ∼ ΔCDB (AA) ⇒ AD/CD = CD/DB ⇒ $CD^2 = AD \cdot DB$ ∎. Numbers: $CD^2 = 3.6 \times 6.4 = 23.04$ ⇒ **CD = 4.8**. AB = 10. $AC^2 = 3.6 \times 10 = 36$ ⇒ **AC = 6**. $BC^2 = 6.4 \times 10 = 64$ ⇒ **BC = 8**.

Part D — Answers

D1.1 P/Q: 9/6 = 1.5, 12/8 = 1.5, 15/10.5 ≈ 1.43 ⇒ **not similar**. P/R: 9/7.5 = 12/10 = 15/12.5 = 1.2 ⇒ **R is similar to P**.

D1.2 The gain is 90 m per 1.5 km = 60 m per km. Total gain = 60 × 4 = **240 m**; top elevation = **560 m**. Assumption: the slope stays constant (similar triangles along the trail).

D1.3 Above 1.8 m, the rise is 1.2 m over 4 m. Similar triangles: rise to B / 10 = 1.2/4 ⇒ 3 m. Post at B = 1.8 + 3 = **4.8 m**.

D1.4 15/6 = (4 + d)/4 ⇒ 4 + d = 10 ⇒ **d = 6**.

D1.5 $k^2 = 12/75 = 0.16$ ⇒ k = 0.4. Small side = 10 × 0.4 = **4 cm**.

D1.6 ΔABC ∼ ΔDEC ⇒ AC/DC = BC/EC: (x + 2)/(3x) = 4/9 ⇒ 9x + 18 = 12x ⇒ **x = 6**, **AC = 8**, **CD = 18**.

D2.1 2.5 × 1.609 = 4.0225 km = **4022.5 m**.

D2.2 3 ÷ 2.47 ≈ **1.21 ha** ≈ **12 146 m²**. (Using 4840 yd² per acre gives ≈ 12 141 m²; the small difference comes from rounding in the table.)

D2.3 0.000045×10^{12} m = 4.5×10^7 m = $\mathbf{4.5 \times 10^4}$ **km**.

D2.4 (a) 2 g = 2000 mg ÷ 4 = 500 mg per dose ⇒ **2 capsules per dose, 8 per day**. (b) 1000 mL ÷ 480 min ≈ **2.08 mL/min**.

D2.5 $30.48^3 \approx 28\,316.8$ cm³ ≈ **28.3 L**.

D3.1 (a) $4\pi(4.5^2) = \mathbf{81\pi \approx 254.5\ cm^2}$. (b) $4\pi r^2 = 196\pi \Rightarrow r^2 = 49 \Rightarrow \mathbf{r = 7\ cm}$.

D3.2 $2\pi(9) + 2\pi(3)h = 48\pi \Rightarrow 18 + 6h = 48 \Rightarrow \mathbf{h = 5\ cm}$.

D3.3 $\pi r(14) = 220 \Rightarrow r = 220/(14\pi) \approx \mathbf{5.0\ cm}$.

D3.4 Slant = $\sqrt{(6^2 + 8^2)} = 10$. Cone $\pi(6)(10) = 60\pi$; cylinder side $2\pi(6)(4) = 48\pi$; base 36π. Total = $\mathbf{144\pi \approx 452.4\ cm^2}$.

D3.5 Pyramid height = 9 − 6 = 3. Slant = $\sqrt{(3^2 + 4^2)} = 5$. Walls 4(8)(6) = 192; base 64; triangles $4 \cdot \frac{1}{2} \cdot 8 \cdot 5 = 80$. Total = **336 cm²**.

D3.6 (a) Sloped bottom: length $\sqrt{(10^2 + 1.5^2)} \approx 10.11$ m × 4 ≈ 40.45 m². Two trapezoid sides: 2 × (1 + 2.5)/2 × 10 = 35 m². Ends: 1 × 4 + 2.5 × 4 = 14 m². Total ≈ **89.45 m²**. (b) 89.45 × 40 × 1.38 ≈ **$4937.50 CAD**.

D4.1 Triangle area = ½(12)(5) = 30 cm². Rectangle area = 6 × 5 = 30 cm². Same cross-sectional area along the same 10 cm length ⇒ both volumes = **300 cm³**. Carl is **wrong**: the area must match, not the shape.

D4.2 $\frac{1}{3}\pi r^2(9) = 150 \Rightarrow r^2 = 450/(9\pi) \approx 15.92 \Rightarrow \mathbf{r \approx 3.99\ cm}$.

D4.3 $\frac{1}{3}(64)h = 256 \Rightarrow \mathbf{h = 12\ m}$. Slant = $\sqrt{(12^2 + 4^2)} = \sqrt{160} \approx \mathbf{12.65\ m}$.

D4.4 Hole = $(\sqrt{3}/4)(4^2)(10) \approx 69.28$. Remaining ≈ 1000 − 69.28 = **930.72 cm³**.

D4.5 Cuboid = 500 ⇒ removed = $100 = \frac{1}{4}\pi r^2(5) \Rightarrow r^2 = 80/\pi \approx 25.46 \Rightarrow \mathbf{r \approx 5.05\ cm}$.

D4.6 Side face: $10 \times 1 + (1 + 3)/2 \times 15 = 10 + 30 = 40$ m². V = 40 × 8 = 320 m³ = **320 000 L**.

D4.7 Box 5 × 2.5 × 8 = 100; roof ½ × 5 × 1.5 × 8 = 30. Total = **130 m³**.

D4.8 $240 + \frac{1}{2}\pi(3^2)(10) = 240 + 45\pi \approx \mathbf{381.37\ cm^3}$.

Part E — Answers

E1 (a) H/26 = 1.8/1.2 ⇒ **H = 39 m**. (b) Now height = shadow (39 m : 39 m), so the pole's shadow = **1.8 m**.

E2 r = 5 × 9/15 = 3 ft. (a) $\pi(3^2) = \mathbf{9\pi \approx 28.27\ ft^2}$. (b) $\frac{1}{3}\pi(9)(9) = \mathbf{27\pi \approx 84.82\ ft^3}$.

E3 k = 15/6 = 2.5 ⇒ $k^2 = 6.25$. Area = 20 × 6.25 = **125 cm²**.

E4 (a) DE/AB = 10/6 = 5/3 and DF/AC = 15/9 = 5/3, and the included angles are equal (∠A = ∠D = 50°) ⇒ **ΔABC ~ ΔDEF (SAS)**. (b) EF = 7 × 5/3 ≈ **11.67**.

E5 $h = \sqrt{(17^2 - 8^2)} = \mathbf{15\ cm}$. $V = \frac{1}{3}\pi(64)(15) = \mathbf{320\pi \approx 1005.3\ cm^3}$. $SA = \pi(8)(17) + 64\pi = \mathbf{200\pi \approx 628.3\ cm^2}$.